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Sysmon and process-creation analysis

Sysmon Event ID 1 records process creation. Depending on configuration and version, fields include image path, command line, parent image and command line, user, hashes, process GUID, and timestamps. Process GUIDs help correlate activity despite PID reuse. Parent-child relationships can reveal unusual execution chains, such as an office application starting a script interpreter, but unusual ancestry is a lead rather than proof of maliciousness.

For PowerShell, examine full command line and parent context for encoded commands, hidden or non-interactive modes, download or web-request behavior, execution-policy bypass flags, and suspicious script paths. These indicators are not conclusive: administrators and software may use similar options. Preserve original logs and correlate with script block logging, endpoint telemetry, network records, and known-good baselines where available.

Limitations: Sysmon must be installed and configured; exclusions, gaps, tampering, retention, and incomplete command-line capture affect conclusions. Validate timestamps and collection context.

Sources: Microsoft Sysmon documentation, https://learn.microsoft.com/sysinternals/downloads/sysmon; Microsoft PowerShell logging documentation, https://learn.microsoft.com/powershell/module/microsoft.powershell.core/about/about_logging.